

Poker_table_HUD

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1 Graphical Interface Design (Lo-Fi Wireframing & HUD)

As part of the Front-End prototyping process, the task of designing the *Heads-Up Display* (HUD) required an iterative approach. The objective was to design a user interface (UI) capable of accommodating the complex rules of Texas Hold'em and the innovative mechanics of quantum betting, while anticipating accessibility constraints for a future mobile port.

1.1 Objectives and Ergonomic Constraints (UX/UI)

The main challenge of the poker table view lay in cognitive overload. The player must simultaneously analyze the board (community cards), monitor the financial status of other players, and manage their own actions within a limited time.

To address these constraints, the interface was built around two guiding principles:

- **Absolute board readability:** The center of the table (containing the main pot and the community cards) must never be obstructed by overlapping menus or interaction pop-ups.
- **Touch accessibility (Mobile-Ready):** The main controls (Fold, Call, Raise) must be grouped in an area easily reachable by the user's thumbs.

1.2 Spatial Architecture and Segregation (BorderPane)

To translate these principles into a JavaFX structure, the spatial architecture relies on the `BorderPane` component, enabling a strict separation between the public game space and the private control space.

1.2.1 The Private Space: The Dashboard (Zone Bottom)

All player interaction logic was extracted from the table and anchored in a static bottom bar. This dashboard includes:

- An enlarged display of the private cards (*Hole Cards*) for immediate readability.
- A horizontal container (**HBox**) aligning the action buttons with standardized color coding (Red for folding, Blue for calling, Green for aggression).
- A dynamic visual feedback badge ("YOUR TURN"), highly contrasted, in order to capture the player's attention during critical phases without cluttering the overall visual space.

1.2.2 The Public Space: The Game Table (*Zone Center*)

The central area, modeled using an **AnchorPane**, is exclusively dedicated to public information. The central board (Flop, Turn, River) features textual labels beneath each card to guide beginner players through the flow of the game.

1.3 Dynamic Player Distribution

To ensure a *responsive* interface (adaptable to different screen sizes), opponent placement is not static. The graphical entities representing players (including their username, chip stack, and card status) are mathematically distributed along an elliptical trajectory within the **AnchorPane**. This trigonometric composition ensures that the distance between the table and the avatars remains proportional when the window is resized.

1.4 Integration of Quantum Innovation (*Hidden Bets*)

The introduction of hidden bets required the creation of a non-intrusive access point. A dedicated button ("Hidden Bets") was integrated on the right lateral edge of the interface. Its purple color (breaking with traditional poker color coding) emphasizes its asynchronous and innovative nature. Its activation will trigger a sliding side panel or a modal window, allowing the consultation of multipliers (*Odds*) without losing sight of the evolution of the central board.



Figure 1: High-Fidelity Wireframe of the HUD and Poker Table (V4)